

A research report on: IoT in real life Smart Agriculture

A smart agriculture system is designed with the use of smart technologies, which saves human effort, time and is more sustainable. It uses Sensors, Artificial intelligence, IoT, GPS, & Data analytics to improve farming efficiency, productivity, and sustainability.

Smart agriculture helps farmers make better decisions by collecting and analysing real-time information about crops, soil, weather, water, and livestock. Instead of treating an entire field the same way, farmers can manage specific areas according to their needs.

Technologies used:

1) Internet of Things (IoT)

- Soil moisture sensors
- Temperature sensors
- Humidity sensors
- Water-level sensors

2) Drones

- Monitor crop health
- Detect pests and diseases
- Assess irrigation needs

3) Map fields, GPS, and GIS

- GPS helps guide tractors and machinery accurately.
- GIS (Geographic Information Systems) helps analyse field data and create digital farm maps.

4) Artificial Intelligence (AI)

- Predict crop diseases
- Recommend fertiliser usage

Reference: Qazi, S., Khawaja, B. A., & Farooq, Q. U., "IoT-Equipped and AI-Enabled Next Generation Smart Agriculture: A Critical Review, Current Challenges and Future Trends," *IEEE Access*, vol. 10, pp. 21219–21235, 2022, doi: 10.1109/ACCESS.2022.3152544.